

IEDA 4000H

Tutorial 6

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- ① More Hints for Homework 2, Problem 2
- ② (Semi) Positive/Negative Definite Matrix

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- For θ : use convexity.
- For f : consider the directional derivative.

- 1 More Hints for Homework 2, Problem 2
- 2 (Semi) Positive/Negative Definite Matrix

Definition

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is said to be

- positive definite if $\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0, \mathbf{x}^T A \mathbf{x} > 0$;
- positive semi-definite if $\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0, \mathbf{x}^T A \mathbf{x} \geq 0$;
- negative definite if $\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0, \mathbf{x}^T A \mathbf{x} < 0$;
- negative semi-definite if $\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq 0, \mathbf{x}^T A \mathbf{x} \leq 0$.

Criterion for Positive/Negative Definiteness

- Eigenvalue criterion: A is positive definite (semi-definite) if and only if all its eigenvalues are positive (non-negative); A is negative definite (semi-definite) if and only if all its eigenvalues are negative (non-positive).
- Principal minor criterion: A is positive definite if and only if all its leading principal minors are positive; A is negative definite if and only if its leading principal minors alternate in sign, i.e., the k -th order leading principal minor (Δ_k) is positive when k is even and negative when k is odd.

Special Case for 2×2 Matrix

For $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, we have

- A is positive definite if and only if $a > 0$ and $ac - b^2 > 0$;
- A is negative definite if and only if $a < 0$ and $ac - b^2 > 0$;
- A is positive semi-definite if and only if $a \geq 0$ and $ac - b^2 \geq 0$;
- A is negative semi-definite if and only if $a \leq 0$ and $ac - b^2 \geq 0$.

Exercises

- (1) Determine if $A = \begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix}$ is positive definite.
- (2) Check if $B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$ is positive or negative definite (or neither).
- (3) For $C = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}$, verify positive definiteness.
- (4) Is $D = \begin{pmatrix} -1 & 0 \\ 0 & -2 \end{pmatrix}$ negative definite? Why?
- (5) Compute the eigenvalues of $E = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ and classify it.

Solutions

- (1) $A > 0$. Eigenvalues: $\lambda_{1,2} = \frac{6 \pm \sqrt{8}}{2} > 0$.
- (2) Neither. Eigenvalues: $\lambda_1 = 3 > 0$ and $\lambda_2 = -1 < 0$ (indefinite).
- (3) Yes. $\Delta_1 = 3 > 0$, $\Delta_2 = 8 > 0$, $\Delta_3 = 22 > 0$.
- (4) Yes. All eigenvalues negative.
- (5) Eigenvalues: $\lambda_1 = 9 > 0$ and $\lambda_2 = 1 > 0$, so positive definite.

Thank you for listening !

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